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Internship Domain: Data Science
Task Number: 04

Technologies Used:

Python, Pandas, SQL, Power BI and google colab

Dataset Description:

The Supermart Grocery Sales dataset consists of 9,994 transaction records containing details about customers, products, categories, sub-categories, cities, regions, order dates, sales, discounts, and profit. The data spans four years, from 2015 to 2018, and provides a useful foundation for evaluating sales trends, customer purchasing patterns, product performance, and overall profitability.

Dataset Source:

<https://www.kaggle.com/datasets/mohamedharris/supermart-grocery-sales-retail-analytics-dataset?resource=download>

Business Insights

1. The dataset generated total sales of 14,956,982 and total profit of 3,747,121.20, resulting in an overall profit margin of approximately 25.05%.
2. Annual sales showed a consistent upward trend, increasing from approximately 2,975,599 in 2015 to 4,977,512 in 2018.
3. In 2018, total sales grew by approximately 28.55% compared with 2017, while total profit increased by around 30.52%, indicating strong business growth.
4. Q4 was the best-performing quarter, contributing approximately 5,525,289 in sales and 1,395,461.39 in profit.
5. November recorded the highest monthly sales at approximately 2,193,924, whereas February had the lowest monthly sales at around 456,102.
6. Eggs, Meat & Fish was the top-performing category in terms of sales, generating approximately 2,267,401.
7. Snacks produced the highest category-level profit of approximately 568,178.85 and achieved the strongest category profit margin of about 25.39%.
8. Oil & Masala showed the weakest overall category performance, with the lowest sales and profit and a profit margin of approximately 24.43%.
9. Health Drinks was the highest-selling sub-category, generating approximately 1,051,439 in sales.
10. Noodles achieved the highest sub-category profit margin at approximately 26.34%, indicating strong profitability.

11. Spices had the lowest sub-category profit margin at approximately 23.82%, suggesting an opportunity for cost or pricing optimization.
12. The West region was the strongest-performing region, recording approximately 4,798,743 in sales and 1,192,004.61 in profit.
13. The North region contains only one transaction record, so its performance is not sufficient for a reliable comparison with the other regions.
14. Kanyakumari generated the highest city-level sales at approximately 706,764, while Vellore recorded the highest city-level profit of around 174,073.01.
15. Krithika was the highest-sales customer, contributing approximately 334,361 in sales, while Arutra generated the highest customer profit of approximately 87,572.40.

ML Forecasting Insight

16. A Holt-Winters Exponential Smoothing model was developed to predict future monthly sales using historical sales patterns, including trend and yearly seasonality. The model was evaluated using the most recent 12 months as test data and achieved a MAE of 44,972.95, RMSE of 58,276.60, and MAPE of 10.80%. After evaluation, the model was retrained using the complete historical dataset and used to forecast sales for the next six months.

Business Recommendations

1. Maintain sufficient inventory for Eggs, Meat & Fish because it is the leading category in terms of sales.
2. Increase promotional visibility for Snacks, as it combines strong sales performance with the highest category profit.
3. Consider expanding the product range and marketing activities for Health Drinks due to its strong sub-category sales.
4. Analyze the pricing, costs, and product mix of Oil & Masala to improve its relatively weak sales, profit, and profit margin.
5. Continue supporting Noodles because its high profit margin indicates strong profitability potential.
6. Review the costs and pricing strategy for Spices to identify opportunities to improve its lowest sub-category profit margin.
7. Prepare additional inventory and staffing capacity for Q4, particularly during November and December, when sales demand is significantly higher.
8. Use the West region as a benchmark by analyzing its successful products and sales patterns and applying relevant strategies to other regions.

9. Investigate the limited data available for the North region before making strategic decisions, as a single record does not provide enough evidence for reliable regional analysis.
10. Avoid relying on discounts as the primary method for increasing profitability. Instead, evaluate discount campaigns at the product or category level to determine their actual impact on profit.

ML-Based Recommendation

11. Use the Holt-Winters six-month sales forecast as an additional planning tool for inventory management, workforce scheduling, and promotional activities. The forecast should be updated regularly as new sales records become available.

Machine Learning Model Used

Holt-Winters Exponential Smoothing

The model uses additive trend and 12-month seasonal patterns to capture changes in monthly sales. The last 12 months of historical sales were reserved for model evaluation, while the remaining data was used for training. The evaluated model achieved a MAPE of 10.80%, indicating an average forecasting error of approximately 10.8%. The final model was then trained on the complete historical dataset and used to generate a six-month future sales forecast.

GitHub Repository Link:

<https://github.com/abdulwahab014-CS/Grocery-sales-analytics-and-forecasting>